Student 1 - Answer to Q1

TCP congestion control stops a sender from flooding the network. Besides the receiver's advertised window (rwnd), the sender maintains a congestion window (cwnd) and may have at most min(cwnd, rwnd) unacknowledged bytes in flight. Slow start: cwnd starts at 1 MSS and increases by 1 MSS for every ACK received, so it doubles every RTT, i.e. it grows exponentially, until it reaches the slow start threshold ssthresh. Congestion avoidance: after ssthresh, cwnd increases by about 1 MSS per RTT, which is linear (additive) increase. On a timeout (RTO) the network is badly congested, so ssthresh is set to cwnd/2, cwnd is reset to 1 MSS and the sender returns to slow start. On three duplicate ACKs, fast retransmit resends the lost segment immediately without waiting for the timer. Fast recovery (TCP Reno) then sets ssthresh to cwnd/2 and cwnd to ssthresh + 3 MSS, and the connection continues in congestion avoidance once a new ACK arrives. This additive increase / multiplicative decrease (AIMD) pattern gives stable and fair sharing of bandwidth.